2025.12.23. 多元函数微分学应用.

1. 由方程组确定的隐函数的求导.

① $\begin{cases} xu+yv=0 \\ uv-xy=5 \end{cases}$ 确定了隐函数 $\begin{cases} u=u(x,y) \\ v=v(x,y) \end{cases}$ 求 $(x,y)=(1,-1)$ 时 u_{xx}, v_{xy}.

(i) 对 x 求导

$$\begin{cases} u+xu_x+yv_x=0 \\ u_xv+uv_x-y=0. \end{cases}$$

代入 $(x,y)=(1,-1)$ 知 $(u,v)=(2,2)$

$$\Rightarrow \begin{cases} u_x-v_x=-2 \\ u_x+v_x=-1 \end{cases} \Rightarrow \begin{cases} u_x=-\frac{5}{4} \\ v_x=\frac{3}{4}. \end{cases}$$

(ii) 对 y 求导.

$$\begin{cases} xu_y+v+yv_y=0 \\ u_yv+uv_y-x=0 \end{cases} \Rightarrow \begin{cases} u_y-v_y=-2 \\ u_y+v_y=\frac{1}{2} \end{cases} \Rightarrow \begin{cases} u_y=-\frac{3}{4} \\ u_x=\frac{5}{4}. \end{cases}$$

(iii) 求两次 x 偏导.

$$\begin{cases} 2u_x+xu_{xx}+yv_{xx}=0 \\ u_{xx}v+uv_{xx}+2u_xv_x=0 \end{cases} \Rightarrow \begin{cases} u_{xx}-v_{xx}=5/2 \\ u_{xx}+v_{xx}=15/16 \end{cases} \Rightarrow \begin{cases} u_{xx}=55/32 \\ v_{xx} \text{ 不关心.} \end{cases}$$

(iv) 先对 y 求导, 再对 x 求导.

$$\begin{cases} u_y+xu_{xy}+v_x+yv_{xy}=0 \\ u_{xy}y+u_yv_x+u_xv_y+uv_{xy}-1=0 \end{cases} \Rightarrow \begin{cases} u_{xy}-v_{xy}=0 \\ u_{xy}+v_{xy}=25/16 \end{cases}$$

$$\Rightarrow \begin{cases} u_{xy}=25/32 \\ v_{xy}=25/32. \end{cases}$$

② $\begin{cases} x = \cos\varphi\cos\theta \\ y = \cos\varphi\sin\theta \\ z = \sin\varphi. \end{cases}$ 求 $\frac{\partial z}{\partial x}$

方法1. 将参数方程写成隐函数

$x^2+y^2+z^2=1$. 对 x 求偏导, $2x+0+2z\frac{\partial z}{\partial x}=0$.

其中, $z=z(x,y)$

$\Rightarrow \frac{\partial z}{\partial x} = -\frac{x}{z}$.

方法2. 通过 Chain rule 硬算.

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial \varphi}\frac{\partial \varphi}{\partial x} + \frac{\partial z}{\partial \theta}\frac{\partial \theta}{\partial x} = \cos\varphi\frac{\partial \varphi}{\partial x} + 0 = \cos\varphi\frac{\partial \varphi}{\partial x}$$

因此需计算 $\frac{\partial \varphi}{\partial x}$.

我们有: $\frac{\partial(x,y)}{\partial(\theta,\varphi)} = \begin{pmatrix} \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial \varphi} \\ \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial \varphi} \end{pmatrix} = \begin{pmatrix} -\cos\varphi\sin\theta & -\sin\varphi\cos\theta \\ \cos\varphi\cos\theta & -\sin\varphi\sin\theta \end{pmatrix}$.

需要 $\frac{\partial(\theta,\varphi)}{\partial(x,y)} = \begin{pmatrix} \frac{\partial \theta}{\partial x} & \frac{\partial \theta}{\partial y} \\ \frac{\partial \varphi}{\partial x} & \frac{\partial \varphi}{\partial y} \end{pmatrix}$.

[理解1.]. $\frac{\partial(\theta,\varphi)}{\partial(x,y)} = \left(\frac{\partial(x,y)}{\partial(\theta,\varphi)}\right)^{-1}$. 用 $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

即 $\frac{\partial(\theta,\varphi)}{\partial(x,y)} \cdot \frac{\partial(x,y)}{\partial(\theta,\varphi)} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

和 $\frac{\partial \varphi}{\partial x} = -\frac{\cos\varphi\cos\theta}{\cos\varphi\sin\varphi} = -\frac{\cos\theta}{\sin\varphi}$

$\Rightarrow \frac{\partial z}{\partial x} = -\frac{\cos\varphi\cos\theta}{\sin\varphi} = -\frac{x}{z}$.

[理解2.].

$$\begin{cases} 1 = \frac{\partial x}{\partial x} = \frac{\partial x}{\partial \theta}\frac{\partial \theta}{\partial x} + \frac{\partial x}{\partial \varphi}\frac{\partial \varphi}{\partial x} = -\cos\varphi\sin\theta\cdot\frac{\partial \theta}{\partial x} - \sin\varphi\cos\theta\cdot\frac{\partial \varphi}{\partial x} \\ 0 = \frac{\partial y}{\partial x} = \frac{\partial y}{\partial \theta}\frac{\partial \theta}{\partial x} + \frac{\partial y}{\partial \varphi}\frac{\partial \varphi}{\partial x} = \cos\varphi\cos\theta\cdot\frac{\partial \theta}{\partial x} - \sin\varphi\sin\theta\cdot\frac{\partial \varphi}{\partial x} \\ 0 = \frac{\partial x}{\partial y} = \frac{\partial x}{\partial \theta}\frac{\partial \theta}{\partial y} + \frac{\partial x}{\partial \varphi}\frac{\partial \varphi}{\partial y} = -\cos\varphi\sin\theta\cdot\frac{\partial \theta}{\partial y} - \sin\varphi\cos\theta\cdot\frac{\partial \varphi}{\partial y} \\ 1 = \frac{\partial y}{\partial y} = \frac{\partial y}{\partial \theta}\frac{\partial \theta}{\partial y} + \frac{\partial x}{\partial \varphi}\frac{\partial \varphi}{\partial x} = \cos\varphi\cos\theta\cdot\frac{\partial \theta}{\partial y} - \sin\varphi\cos\theta\cdot\frac{\partial \varphi}{\partial x} \end{cases}$$

解四元一次方程 $\Rightarrow \frac{\partial \varphi}{\partial x} = -\frac{\cos\theta}{\sin\varphi}$. ✓.

③ $x^2\frac{\partial z}{\partial x}+y^2\frac{\partial z}{\partial y}=z^2$.

$x=t,\ y=\frac{t}{1+tu},\ z=\frac{t}{1+tw}$，求证 $\frac{\partial w}{\partial t}=0$.

Step1. 写变量之间关系（计算图）☆

$$\begin{tikzcd}
(x,y) \arrow[r] \arrow[d, leftrightarrow] & z \arrow[r] & w. \\
(u,t) \arrow[ur] \arrow[ur, bend right] & t \arrow[ur] &
\end{tikzcd}$$

Step2. Chain rule.

$W=\frac{1}{z}-\frac{1}{t}$. $\Rightarrow \frac{\partial w}{\partial t}=\frac{1}{t^2}-\frac{1}{z^2}\frac{\partial z}{\partial t}$.

因此只需验证 $\frac{\partial z}{\partial t}=\frac{z^2}{t^2}$

$\frac{\partial z}{\partial t}=\frac{\partial x}{\partial t}\frac{\partial z}{\partial x}+\frac{\partial y}{\partial t}\frac{\partial z}{\partial y}$ 其中，$\frac{\partial x}{\partial t}=1$，$\frac{\partial y}{\partial t}=\frac{1}{(1+tu)^2}=\frac{y^2}{x^2}$.

$=\frac{\partial z}{\partial x}+\frac{y^2}{x^2}\cdot\frac{\partial z}{\partial y}=\frac{z^2}{x^2}=\frac{z^2}{t^2}$. □.

2. 条件极值问题（Lagrange 乘子法）.

[方法论].

Step1. 将问题写成 $\min f(x,y,z)$

s.t. $\begin{cases} g(x,y,z)=0 \\ h(x,y,z)=0 \end{cases}$ 情形.

可以有多个约束

Step2. 写成 Lagrange 函数

"原函数 + 乘子×约束"

$L(x,y,z,\lambda,\mu)=f(x,y,z)+\lambda g(x,y,z)+\mu h(x,y,z)$

乘子与约束数目相同.

Step3. 求解方程 $\nabla L=0$.

解即全部条件驻点候选.

Eg1. 求曲面 $(x-y)^2-z^2=4$ 到原点距离.

$d=\sqrt{x^2+y^2+z^2}$. 等价地，我们最小化 $x^2+y^2+z^2$.

Step1. $\min\ x^2+y^2+z^2$

s.t. $(x-y)^2-z^2-4=0$

Step2. $L(x,y,z,\lambda)=x^2+y^2+z^2+\lambda\left((x-y)^2-z^2-4\right)$

Step3. $\begin{cases}\frac{\partial L}{\partial x}=2x+2\lambda x-2\lambda y=0\\ \frac{\partial L}{\partial y}=2y+2\lambda y-2\lambda x=0\\ \frac{\partial L}{\partial z}=2z-2\lambda z=0\\ \frac{\partial L}{\partial \lambda}=(x-y)^2-z^2-4=0\end{cases}$ $\Leftrightarrow$ $\begin{cases}\left.\begin{array}{l}(1+\lambda)x=\lambda y\\(1+\lambda)y=\lambda x\end{array}\right\}\Rightarrow x+y=0\\(1-\lambda)z=0\\(x-y)^2-z^2=4.\end{cases}$

(i) $\lambda=1$，则 $\begin{cases}2x=y\\2y=x\end{cases}$ $\Rightarrow x=y=0$ $\Rightarrow z^2=-4$ ✗

故 (ii) $\begin{cases}z=0\\x+y=0\\x-y=\pm2\end{cases}$ $\Rightarrow (x,y,z)=(1,-1,0)$ or $(-1,1,0)$

$\Rightarrow d=\sqrt{2}$ 为最小 ✓.

Eg. (2021.7). 求单位圆内接 n 边形面积最大值.

用 n 条边所对圆心角 $\theta_1\sim\theta_n$ 表示.

$S=f(\theta_1,\dots,\theta_n)=\sum_{k=1}^{n}\frac{1}{2}\sin\theta_k$.

约束：$\theta_1+\dots+\theta_n-2\pi=0$

Step1. $\min\ \frac{1}{2}\sum_{k=1}^{n}\sin\theta_k$

s.t. $\sum_{k=1}^{n}\theta_k-2\pi=0$.

Step2. $L(\theta_1,\dots,\theta_n,\lambda)$

$=\frac{1}{2}\sum_{k=1}^{n}\sin\theta_k+\lambda\left(\sum_{k=1}^{n}\theta_k-2\pi\right)$.

Step3. $\begin{cases}\frac{\partial L}{\partial\theta_i}=\frac{1}{2}\cos\theta_k+\lambda=0\\ \frac{\partial L}{\partial\lambda}=\sum_{k=1}^{n}\theta_k-2\pi=0\end{cases}$ $\rightsquigarrow$ $i\neq j$ 时，$\cos\theta_i=\cos\theta_j$

$\Leftrightarrow$ $\theta_i=\theta_j$ 或 $2\pi-\theta_j$

但 $\sum_{k=1}^{n}\theta_k=2\pi$，$n\geq3$

故只能 $\theta_i=\theta_j$. $\forall i\neq j$.

$\Rightarrow S_{max}=f\left(\frac{2\pi}{n},\dots,\frac{2\pi}{n}\right)=\frac{n}{2}\sin\frac{2\pi}{n}$.

3. Morse型问题.

Eg.1. (2024.6).

$f(x,y) = x^2+2xy\sin(x+y)-y^2$. 求证：在 $(0,0)$ 附近存在一个可逆可微映射 $\begin{cases} u=u(x,y) \\ v=v(x,y)\end{cases}$ s.t. $u(0,0)=v(0,0)=0$. $f(x(u,v),y(u,v)) = u^2-v^2$.

分析：可逆性是用逆映射定理，即说明 $\left|\frac{\partial(u,v)}{\partial(x,y)}\right| \neq 0$.

因此需构造出 $u(x,y)$, $v(x,y)$

解：配方. $f(x,y) = \left(x^2+2xy\sin(x+y)+y^2\sin^2(x+y)\right) - y^2\left(1+\sin^2(x+y)\right)$

$$= \underbrace{\left(x+y\cdot\sin(x+y)\right)}_{u(x,y)}{}^2 - \underbrace{\left(\sqrt{1+\sin^2(x+y)}\cdot y\right)}_{v(x,y)}{}^2.$$

$\cancel{\nabla f = \begin{pmatrix} 2x+2y\sin(x+y)+2xy\cos(x+y) \\ 2x\sin(x+y)\end{pmatrix}}$.

$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y}\end{pmatrix} = \begin{pmatrix} 1+y\cos(x+y) & \sin(x+y)+y\cos(x+y) \\ y\cdot\frac{2\sin(x+y)\cos(x+y)}{\sqrt{1+\sin^2(x+y)}} & y\cdot\frac{2\sin(x+y)\cos(x+y)}{\sqrt{1+\sin^2(x+y)}}+\sqrt{1+\sin^2(x+y)}\end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1\end{pmatrix} \text{（在 } (0,0) \text{ 处）}.$$

$\left|\frac{\partial(u,v)}{\partial(x,y)}\right| \neq 0$. 因此可逆. □.

Eg.2. (2022.9).

f 在 $(0,0)$ 附近满足 $f(0,0)=0$, $\nabla f\begin{pmatrix}0\\0\end{pmatrix} = \begin{pmatrix}0\\0\end{pmatrix}$, $\nabla^2 f = \begin{pmatrix}\alpha & \beta \\ \beta & \gamma\end{pmatrix}$, $\alpha>0$, $|\nabla^2 f|<0$.

则 (1) $f(x,y) = x\cdot a(x,y) + y\cdot b(x,y)$. $a(0,0)=b(0,0)=0$.

(2) 与Eg1. 相同.

(1). $f(x,y) = \underline{\int_0^1 \frac{d}{dt} f(tx,ty)\,dt + f(0,0)}$ (N-L公式)

$= \int_0^1 \left(x\cdot f_1(tx,ty) + y\cdot f_2(tx,ty)\right) dt$

$= x\cdot \underbrace{\int_0^1 f_1(tx,ty)\,dt}_{a(x,y)} + y\cdot \underbrace{\int_0^1 f_2(tx,ty)\,dt}_{b(x,y)}$

$= x\cdot a(x,y) + y\cdot b(x,y)$

Rmk. 以上仅用到 $f(0,0)=0$! 明显还知道 a, b 可微 若 f_1, f_2 连续.

由于 $\nabla f(0,0)=0$. 我们计算 $\nabla f = \begin{pmatrix} a(x,y) + x\cdot a_1(x,y) + y\cdot b_1(x,y) \\ x\cdot a_2(x,y) + b(x,y) + y\cdot b_2(x,y) \end{pmatrix}$

在 $(0,0)$ 处为 $\begin{pmatrix} a(0,0) \\ b(0,0) \end{pmatrix}$.

因此 $a(0,0) = b(0,0) = 0$. $\square$.

(2) 与(1)同理, 由于 $a(0,0) = b(0,0) = 0$

知 $a(x,y) = x\cdot A(x,y) + y\cdot B(x,y)$

$b(x,y) = x\cdot C(x,y) + y\cdot D(x,y)$

Rmk. 由于 $\nabla a(0,0)$, $\nabla b(0,0)$ 不是0 ~~知道~~, 因此 $A(0,0)$, $B(0,0)$, $C(0,0)$, $D(0,0)$ 不~~知道~~是 0.

$\Rightarrow f(x,y) = A(x,y)\cdot x^2 + (B(x,y) + C(x,y))\cdot xy + D(x,y)\cdot y^2$

总之形如 $f(x,y) = x^2\cdot F(x,y) + 2xy\cdot G(x,y) + y^2\cdot H(x,y)$.

$$\nabla f = \begin{pmatrix} 2x + x^2F_x + 2yG + 2xyG_x + y^2H_x \\ x^2F_y + 2xG + 2xyG_y + 2yH + y^2H_y \end{pmatrix}.$$

$$\nabla^2 f = \begin{pmatrix} 4xF_x + 2F + x^2F_{xx} + 4yG_x + 2xyG_{xx} + y^2H_{xx} & \underline{2xF_y + x^2F_{xy} + 2G + 2yG_y + 2xG_x + 2xyG_{xy} + 2yH_y + y^2H_{xy}} \\ \text{这俩一样.} & x^2F_{yy} + 4xG_y + 2xyG_{yy} + 2H + 4yH_y + y^2H_{yy} \end{pmatrix}$$

$\Rightarrow \nabla^2 f(0,0) = \begin{pmatrix} 2F(0,0) & 2G(0,0) \\ 2G(0,0) & 2H(0,0) \end{pmatrix} \overset{\text{条件}}{=\!=} \begin{pmatrix} \alpha & \beta \\ \beta & \gamma \end{pmatrix}$. 知 $F(0,0) > 0$, $F(0,0)H(0,0) < G^2(0,0)$

由连续性, 在 $(0,0)$ 附近, $F(x,y) > 0$, $F(x,y)H(x,y) < G^2(x,y)$.

因此，可以配方.

$$f(x,y)=\left(F(x,y)\cdot x^2+2xy\cdot G(x,y)+\frac{G^2(x,y)}{F(x,y)}\cdot y^2\right)+y^2\cdot\left(H(x,y)-\frac{G^2(x,y)}{F(x,y)}\right)$$

$$=\underbrace{\left(\sqrt{\underbrace{F(x,y)}_{>0}}\left(x+\frac{G(x,y)}{F(x,y)}\cdot y\right)\right)^2}_{u(x,y)}-\underbrace{\left(\sqrt{\underbrace{\frac{G^2(x,y)}{F(x,y)}-H(x,y)}_{>0}}\cdot y\right)^2}_{v(x,y)}$$

$$\frac{\partial(u,v)}{\partial(x,y)}=\begin{pmatrix}\frac{F_x}{2\sqrt{F}}\left(x+\underbrace{\frac{G}{F}y}_{=0}\right)+\sqrt{F}\left(1+\frac{G_xF-GF_x}{F^2}\underbrace{y}_{=0}\right) & \cancel{\sqrt{F}}\ \frac{F_y}{2\sqrt{F}}\left(x+\underbrace{\frac{G}{F}y}_{=0}\right)+\sqrt{F}\left(\frac{G}{F}+\underbrace{y}_{=0}\cdot\frac{G_yF-GF_y}{F^2}\right)\\ \underbrace{y}_{=0}\cdot\frac{\partial}{\partial x}\sqrt{\frac{G^2}{F}-H} & \sqrt{\frac{G^2}{F}-H}+\underbrace{y}_{=0}\cdot\frac{\partial}{\partial y}\sqrt{\frac{G^2}{F}-H}\end{pmatrix}$$

在 $(0,0)$ 处 $=\begin{pmatrix}\underbrace{\sqrt{F(0,0)}}_{>0} & G(0,0)/\sqrt{F(0,0)}\\ 0 & \underbrace{\sqrt{\frac{G^2(0,0)}{F(0,0)}-H(0,0)}}_{>0}\end{pmatrix}$ $\Rightarrow \left.\left|\frac{\partial(u,v)}{\partial(x,y)}\right|\right|_{(0,0)}\neq 0$

知可逆性. □

4. L'Hopital 法则与中值定理拾遗

① 证明2个函数相等. $f(x)=g(x)$

$$\left.\begin{array}{l}\text{Step1. } \exists x_0 \text{ s.t. } f(x_0)=g(x_0)\\ \text{Step2. } f'(x)=g'(x)\end{array}\right\}\ f(x)=g(x).$$

Eg. (2023.8).

(1) $\left(-\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right)$ 上 $2\int_0^x\frac{dt}{\sqrt{1-t^2}}=\int_0^{2x\sqrt{1-x^2}}\frac{dt}{\sqrt{1-t^2}}$

(2) $\left(-\frac{1}{\sqrt[4]{6}},\frac{1}{\sqrt[4]{6}}\right)$ 上, $2\int_0^x\frac{dt}{\sqrt{1-t^4}}=\int_0^{\frac{2x\sqrt{1-x^4}}{1+x^4}}\frac{dt}{\sqrt{1-t^4}}$

证明: $x=0$ 时, 两端相等

(1) $(LHS)'=\frac{2}{\sqrt{1-x^2}}$

$$(RHS)' = \frac{1}{\sqrt{1-4x^2(1-x^2)}} \cdot \left(2\sqrt{1-x^2} - 2x \cdot \frac{x}{\sqrt{1-x^2}} \right)$$

$$= \frac{1}{\sqrt{(2x^2-1)^2}} \cdot \frac{2}{\sqrt{1-x^2}} \left(1-2x^2 \right) = \frac{1}{1-2x^2} \cdot \frac{2}{\sqrt{1-x^2}} \cdot (1-2x^2)$$

$$= \frac{2}{\sqrt{1-x^2}}. \quad \square.$$

(2) $(LHS)' = \dfrac{2}{\sqrt{1-x^4}}$

$$(RHS)' = \left(\sqrt{1 - \frac{16x^4(1-x^4)^2}{(1+x^4)^4}} \right)^{-1} \cdot \frac{1}{(1+x^4)^2} \left(2\sqrt{1-x^4}(1+x^4) + 2x \cdot \frac{-4x^3}{2\sqrt{1-x^4}}(1+x^4) - 4x^3 \cdot 2x\sqrt{1-x^4} \right)$$

$$= \left(\frac{1}{(1+x^4)^2} \sqrt{\underbrace{x^{16} - 12x^{12} + 38x^8 - 12x^4 + 1}_{(x^8-6x^4+1)^2}} \right)^{-1} \cdot \frac{1}{(1+x^4)^2} \cdot \frac{2}{\sqrt{1-x^4}} \cdot (x^8 - 6x^4 + 1)$$

$$= \frac{2}{\sqrt{1-x^4}}. \quad \square.$$

② 带积分的极限.

适合用 L'Hopital, 因为会将积分号去掉!

Eg. (2023.1).

$$\lim_{x\to 0} \frac{\sin\left(x - \int_0^x \sqrt{1+t^2}\,dt\right)}{x^3} \overset{L'H}{=\!=} \lim_{x\to 0} \frac{\cos\left(x - \int_0^x \sqrt{1+t^2}\,dt\right) \cdot \left(1 - \sqrt{1+x^2}\right)}{3x^2}$$

$$= \lim_{x\to 0} \cos\left(x - \int_0^x \sqrt{1+t^2}\,dt\right) \cdot \left(-\frac{1}{3}\right) \cdot \lim_{x\to 0} \frac{\sqrt{1+x^2} - 1}{x^2}$$

$$= -\frac{1}{3} \lim_{x\to 0} \frac{1 + \frac{x^2}{2} + o(x^2) - 1}{x^2} = -\frac{1}{6}.$$

③ (2023.9).

$P(x)$ 为 $(0,1)$ 上严格↗可导函数 $P(0)=0,\ P(1)=1$. 求证: $\exists\ 0<\theta_0<\theta_1<\cdots<\theta_n<1$

s.t. $(A+B)^n=\sum_{k=0}^{n}\frac{1}{P'(\theta_k)}C_n^k A^{n-k}B^k$.

分析: $\frac{1}{P'(\theta)}$ 形式 一般由反函数求导给出, 先令

$Q(\theta)=P^{-1}(\theta)$.

注意, 考虑 $Q(x)=P^{-1}(x)$. $Q'(x)=\frac{1}{P'(Q(x))}$.

故 $\Leftrightarrow$ 找 $0<x_0<x_1<\cdots<x_n<1$ s.t. $\xi_k=P(x_k)$.

$(A+B)^n=\sum_{k=0}^{n}Q'(\xi_k)C_n^k A^{n-k}B^k$.

由于 x_k,ξ_k 一一对应, 故 $\Leftrightarrow$ 找 $0<\xi_0<\xi_1<\cdots<\xi_n<1$

s.t. $(A+B)^n=\sum_{k=0}^{n}Q'(\xi_k)C_n^k A^{n-k}B^k$.

这里, $Q(0)=0$, $Q(1)=1$, $Q(x)$ 严格↗.

$Q'(\xi)$ 一般由 Lagrange 中值定理构造

在 (y_k, y_{k+1}) 上 $\exists\ \xi_k$ s.t. $Q'(\xi_k)=\frac{Q(y_{k+1})-Q(y_k)}{y_{k+1}-y_k}$.

转为寻找 $0=y_0<y_1<\cdots<y_n<y_{n+1}=1$.

$\xi_0<\xi_1\cdots\xi_{n-1}<\xi_n$

即 $\Leftrightarrow$ 找 $0=y_0<y_1<\cdots<y_n<y_{n+1}=1$ s.t.

$(A+B)^n=\sum_{k=0}^{n}\frac{Q(y_{k+1})-Q(y_k)}{y_{k+1}-y_k}C_n^k A^{n-k}B^k$

$\Leftrightarrow\quad 1=\sum_{k=0}^{n}\left\{\frac{C_n^k A^{n-k}B^k}{(A+B)^n(y_{k+1}-y_k)}\right\}\cdot\left(Q(y_{k+1})-Q(y_k)\right)$

如果上这个 $=1$

RHS 就裂项为 $\sum_{k=0}^{n}(Q(y_{k+1})-Q(y_k))$

$=Q(y_{n+1})-Q(y_0)$

$=Q(1)-Q(0)$

$=1-0=1$ ✓

因此，只需构造 $y_{k+1}-y_k = C_n^k A^{n-k}B^k/(A+B)^n$.

这当然可行！只需取 $\begin{cases} y_0 = 0 \\ y_{k+1} = y_k + C_n^k A^{n-k}B^k/(A+B)^n \end{cases}$

（易验证：$y_{n+1} = \sum_{k=0}^{n} \frac{C_n^k A^{n-k}B^k}{(A+B)^n} = 1$ ）.

即可. $\square$.